V-T-291E
July 1, 1976
SUPERSEDING
Fed. Spec. V-T-291D
January 10, 1966

FEDERAL SPECIFICATION

THREAD, LINEN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for linen thread used for machine and hand sewing.

1.2 Classification.

1.2.1 Types and classes. The thread shall be of the following types and classes as specified (see 6.1).

Type A	-	Three-cord (stitching)
Class 1	-	Natural finish
Class 2	-	Polished finish
Type B	-	Lockstitch twist
Class 1	-	Natural finish
Class 2	-	Polished finish
Type C	-	Hand shoe
Class 1	-	Natural finish

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-50	-	Packaging and Packing of Thread for Domestic and Overseas Shipment

FSC 8310

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Federal Standard:

Fed. Test Method Std No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Dcouments, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Ft. Worth, Denver, San Francisco, Los Angeles and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Color Association of the United States, Inc.

Department of Defense Standard Shades for Sewing Thread 1968

(Application for copies should be addressed to the Color Association of the United States, Inc., 200 Madison Avenue, New York, NY 10016.)

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3. REQUIREMENTS

3.1 Government and supply purchases. The requirements specified in 3.8, 3.9, 3.10, section 5, and related provisions of section 4 apply only to thread purchased directly by the Government. All other requirements apply both to thread purchased by a supplier as a component for an end item and to thread purchased directly by the Government.

3.2 Standard sample. When a standard sample is established, the thread shall match the standard for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.2).

3.3 Material. The thread shall be made of "long-line" flax.

3.4 Color. The thread shall be natural (boiled or scoured), bleached white, or in dyed shades (except type C), as specified in the applicable end item specification or in the contract or order (see 6.1). Shade designations by letter (e.g. Olive Drab S-1) and related cable numbers (e.g. CA 66022) refer to the DOD Standard Shades for Sewing Threads 1968 (see 2.2).

3.4.1 Matching. The shade of the end item threads after removal of finish when applicable, shall match the specified applicable approved shade standard (see 6.2), when tested as specified in 4.2.4.

3.4.2 Sulfur content. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished thread shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.2.4. When a standard sample is not available, the dyed and finished thread shall show not more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.2.4.

3.4.3 Colorfastness. The colorfastness of dyed finished thread shall be as specified in the applicable end item specification or as set forth in the contract or order (see 6.1) except that when requirements are not stated or referenced to a standard sample for colorfastness, the thread shall show "good" fastness to laundering, (after 3 cycles), and light. When a standard sample for colorfastness is referenced, the thread shall be equal to or better than the standard sample for the specified colorfastness properties or (when specific properties are not specified) for colorfastness to laundering, (after 3 cycles) and light. Tests for colorfastness (including tests for wet dry-cleaning and weathering, when specified), shall be as specified in 4.2.4.

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3.5 Finish. The thread shall have a natural or a polished finish as indicated by class and specified in the applicable end item specification or procurement document (see 6.1). The finish shall conform to the applicable requirements of 3.5.1 through 3.5.4.

3.5.1 Class 1. Class 1 thread shall have a natural finish having no consistency or luster other than that inherent in the linen material.

3.5.2 Class 2. Class 2 thread shall be polished and lustrous and shall have a smooth dressed surface with no protruding fibers.

3.5.3 Nonfibrous materials content. Class 1 thread shall have no more than 8.0 percent and the class 2 thread no more than 14.0 percent nonfibrous materials content when tested as specified in 4.2.4.

3.5.4 Finish materials. The thread shall have no chemical finishes or treatments other than those commonly used in commercial threads or those demonstrated to have no deleterious effects on linen thread including effects in prolonged storage. No finish or treatment shall be applied for the purpose of increasing breaking strength.

3.6 Direction of twist. Types A and B threads shall have "S" or "Z" final twist as specified (see 6.1). Type C thread shall have "Z" twist. Direction of twist shall be determined as specified in 4.2.4.

3.7 Physical requirements. The finished thread shall conform to the applicable requirements of tables I, II, and III for number of plies, minimum length per pound, and minimum breaking strength when tested as specified in 4.2.4.

3.7.1 Maximum length per pound. The maximum length per pound of any stated ticket number of thread shall be the specified minimum of the next finer size thread of the same type. The maximum length per pound of the finest size of threads listed for each type shall be not more than 12.5 percent above the minimum specified.

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TABLE I. Type A

Ticket number	Ply	Minimum length per pound (yards) Class 1	Minimum length per pound (yards) Class 2	Minimum breaking strength (pounds) classes 1 and 2
16	3	1,500	1,410	19.0
20	3	1,860	1,748	18.0
25	3	2,320	2,180	12.0
30	3	2,800	2,632	10.0
35	3	3,300	3,102	8.5
40	3	3,800	3,572	7.5
50	3	4,750	4,465	6.0

TABLE II. Type B

Ply	Minimum length per pound (yards) Class 1	Minimum length per pound (yards) Class 2	Minimum breaking strength (pounds) classes 1 and 2
3	1,860	1,780	18
4	1,400	1,300	24
5	1,120	1,070	30
6	930	890	38
7	800	770	44
8	700	670	52
9	620	590	59
10	560	540	65
11	530	510	72
12	485	455	81

TABLE III. Type C

Ticket number	Ply	Minimum length per pound (yards)	Minimum breaking strength (pounds)
3	1	4,060	6
10	1	2,850	10

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3.8 Knots and splices. Types A and C thread shall average not more than one thread knot or splice per 1,000 yards. Type B thread shall average not more than one knot or splice per 500 yards.

3.9 Put-up. Unless otherwise specified, the thread shall be put up on a nominal weight per holder basis on commercial spools, tubes, or balls, (hereinafter referred to as holder), in amounts as specified in the contract or order in accordance with the schedule of table IV (see 6.1). The average net weight per holder shall be not less than the specified nominal weight per holder minus 3.0 percent tolerance when examined (on a 10-holder basis) as specified in 4.2.2. The thread shall be put up in one continuous length and shall be so wound that each turn and layer is free from entanglement.

TABLE IV. Put-ups

Type	Put-up
A	2 ounce spools; 8 or 16 ounce tubes
B	8, 16, 24 or 32 ounce tubes
C	2 ounce balls; 8 or 16 ounce tubes

3.10 Identification marking. Except when acceptance of commercial identification markings are specified in the invitation for bid (see 6.1), each holder shall have a label attached in such a manner as to remain in place and be clearly legible until all thread has been removed. The label shall be printed with the information as specified below:

Stock number
Net weight (oz. or lbs.)
Direction of twist
Color
Ticket number and ply
Type and class
Nomenclature

3.11 Workmanship. The finished thread shall be clean, uniform in color, have no objectional odor, and shall conform to the quality and grade of product established by this specification. The occurence of defects shall not exceed the applicable acceptable quality levels (AOLs).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling and inspection shall be performed in accordance with MIL-STD-105 except where otherwise indicated.

4.2.1 End item examination.

4.2.1.1 Thread on the holder. The thread shall be examined on the holder and the defects listed in table V shall be counted. The sample unit for this examination shall be one wound holder, and the lot size shall be expressed in units of one holder each. The AOL shall be 4.0 defects per 100 units and the inspection level shall be S-3 of MIL-STD-105.

TABLE V. Defects of thread on holders

Examine	Defect
Identification label	Missing, incorrect, incomplete, illegible, or insecurely attached
Type of holder	Other than specified
Surface condition	Loose ply, cut, torn, chafe, slip, affecting strength of thread or interfering with easy location of end and initial unwinding
Color	Other than specified

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TABLE V. Defects of thread on holders (cont'd)

Examine	Defect
Cleanliness	Dirt, spot or stain clearly noticeable
Finish	Other than specified Uneven, lumpy, bare or thin spots (class 2) Not smooth and lustrous (class 2)

4.2.1.2 Thread as unwound. The thread shall be examined by unwinding and defects listed in table VI shall be counted regardless of their proximity to each other. The sample unit for this examination shall be one holder, and the lot size shall be expressed in units of one holder each. The AOL shall be 4.0 defects per 100 units and the inspection level shall be S-3 of MIL-STD-105.

TABLE VI. Defects of thread as unwound

Examine	Defect
Continuous length	Not in continuous length
Knots and splices	Averaging more than specified maximum number of thread knots or splices (types A and C thread one per 1,000 yards; type B, one per 500 yards)
Winding	Improperly or not firmly wound, resulting in kinks, knots, entangling, or slippage during unwinding or otherwise affecting free unwinding of the thread
Tackiness or adhesion	Strands adhere to each other or to holder, affecting unwinding tension

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4.2.2 Examination for weight. The sample unit for this examination shall be 10 wound holders, taken at random from the lot under test. The sample size shall be one sample unit regardless of lot size. The holders shall be weighed individually. The weights of thread obtained shall be averaged and this average shall represent the net weight of thread per holder in the lot. The lot shall be unacceptable if the net weight per holder is less than the amount specified minus the 3.0 percent tolerance (see 3.9).

4.2.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-50 to determine whether packaging, packing, and marking comply with section 5 requirements.

4.2.4 Testing of the end item. The methods of testing specified in FED-STD-191 whereever applicable and as listed in table VII shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The AOL for each characteristic shall be 6.5 test failures per 100 units and the inspection level shall be level S-1. The sample unit shall be one holder. The lot shall be expressed in units of one holder each.

TABLE VII. Test methods

Characteristic	Requirement paragraph	Test method
Long-line flax	3.3	1/
Shade match	3.4	4.2.4.1 and 9010
Presence of labile sulfur	3.4.2	2020
Colorfastness (as applicable) to:		
Laundering (after 3 cycles)	3.4.3	5610 2/ 3/
Light	3.4.3	4660 4/
Wet drycleaning	3.4.3	4622
Weathering	3.4.3	4671 5/

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TABLE VII. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Finish:		
Class 6/	3.5	Visual
Materials limitations	3.5.4	1/
Nonfibrous materials	3.5.3	2620
Direction of final twist	3.6	4050
Number of plies	3.7	Visual 7/
Length per pound	3.7	4010
Breaking strength	3.7	4100

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ The specimens must be dried after each of the three laundering cycles.

3/ On the color transfer cloth evaluation, only the stain on the cotton fibers of the color transfer cloth shall be evaluated.

4/ The time of exposure shall be 40 standard fading hours.

5/ The time of exposure shall be 40 hours.

6/ To be performed in lieu of applicable examination under 4.2.1 when thread is furnished by the supplier as a component.

7/ One determination, and the result reported as "pass" or "fail".

4.2.4.1 Shade evaluation.

4.2.4.1.1 Removal of finish (class 2 only). Before evaluation for shade matching, the sample of class 2 (polished finish) thread shall be wet dry-cleaned in accordance with Method 4622 (disregarding references to the standard sample). Excess solvent shall be removed by centrifuging or wringing.

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The sample shall then be rinsed in distilled water at 120° to 160°F., and dried at a temperature not exceeding 180°F. The dried sample shall then be conditioned for a minimum of four hours prior to evaluation for shade match according to the procedure of method 9010.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A, B or C as specified (see 6.1).

5.1.1 Levels A, B, and C. The thread shall be put-up and packaged in accordance with the applicable requirements of PPP-P-50.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Levels A, B, and C. The thread shall be packed in accordance with the applicable requirements of PPP-P-50.

5.3 Marking. In addition to any special markings required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-50.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, size, and nomenclature of thread required (see 1.2).
(c) Color (shade) required (see 3.4).
(d) Colorfastness properties required (see 3.4.3).
(e) Finish required (see 3.5).
(f) Direction of twist for types A and B (see 3.6).
(g) Put-up required (see 3.9).
(h) When commercial identification markings are acceptable (see 3.10).
(i) Selection of applicable levels of put-up and packaging and packing (see 5.1 and 5.2).
(j) Selection of applicable marking requirement (see 5.3).

6.2 Standard sample. For military shades, the standards for shade reference are those contained in the standard color card for the official standardized shades of sewing threads 1968 (see 2.2 and 3.4). For other shades, address the procuring office issuing the invitation for bids or request for proposal. Standard samples frequently used by the military (directly or for component end items) have been made available to normal suppliers as basic reference for the establishment of the supplier's secondary standards conforming to the characteristics of the official standards.

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The basic shades are those appearing in the DOD Color Card. Prime thread manufacturers who inadvertently have not been furnished standards for establishment of their colorfastness standards in accordance with this specification may obtain samples for this purpose from US Army Natick Research and Development Command, Natick, MA 01760, ATTN: DRXNM-VT.

6.3 Colorfastness guidelines. Normally, colorfastness requirements for thread are in keeping with those specified for the basic material into which the thread will be sewed (see 3.4.3). Typical examples are:

Use	Colorfastness
Thread for footwear	No colorfastness tests apply
Thread for tentage	Good colorfastness to weathering
Thread for equipage	Good colorfastness to laundering and weathering

6.5 Level A packaging and level B packing shall be used by ordering agencies only when shipment is known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

Custodians:

Army - GL
Navy - SA
Air Force - 82

Review activities:

Army - EA, MI, MD
Navy - MC
Air Force - 11, 45
DNA - DS
DSA - CT

User activity:

Army - ME

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

DC
GSA-FSS
HEW

Project No. 8310-0080

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